Citation-Aware Dense Retrieval for Scientific Literature Search

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ABSTRACT

Researchers rarely search for papers with the vocabulary of the papers they need. Lexical rankers such as BM25 reward term overlap and miss work phrased in the conventions of a neighbouring field, while general-purpose dense retrievers are trained on web questions and transfer poorly to technical prose. We present CADRE, a dual-encoder retriever that treats the citation graph as supervision: co-cited papers become positives, and papers that share vocabulary but are never cited together become hard negatives. On three literature-search benchmarks CADRE improves nDCG@10 by 4.1 points over the strongest dense baseline and recovers 91% of the relevant papers in the top 100, at the latency of a single nearest-neighbour lookup.

KEYWORDS dense retrieval, scientific literature, citation graphs, contrastive learning, hard negatives

1 Introduction

A literature search is a recall problem with a precision budget. A reader drafting a related-work section wants every paper that bears on the claim, but will read only the first page of results. Classical rankers built on the probabilistic relevance framework [1, 2] remain strong on this task because scientific queries are full of rare, discriminative terms. They fail when the same idea is named differently: *contrastive pre-training* in one community is *metric learning* in another, and a query in one dialect ranks the other far below the fold.

Dense retrievers map queries and documents into a shared vector space and score them by inner product [3]. Their quality depends almost entirely on the training pairs. Web-scale question–passage pairs teach a model what a good answer to a factoid question looks like, not what a relevant prior study looks like. Scientific text, however, comes with its own relevance judgements: authors cite what they consider related. We exploit this signal directly, building on earlier work that used citation contexts as weak supervision for document embeddings [4].

Our contributions are threefold. First, we define positives and negatives from the citation graph in a way that avoids the false negatives that plague mined hard examples [5]. Second, we combine the graph-derived loss with distillation from a cross-encoder [6]. Third, we evaluate on three benchmarks drawn from twelve fields [7] and report where the model still fails (Section 4).

*Correspondence: m.okafor@example.org. Code and indices will be released on publication.

2 Method

2.1 Dual-encoder scoring

Let q be a query and d a document, represented by its title and abstract. Two encoders f_θ and g_ϕ with shared weights map them to unit vectors in $\mathbb{R}^k$, and the relevance score is their inner product,

$$s(q, d) = \frac{f_\theta(q)^\top g_\phi(d)}{\tau}, \tag{1}$$

where $\tau > 0$ is a learned temperature. Because $g_\phi(d)$ does not depend on the query, every document is encoded once and stored in an approximate nearest-neighbour index [8]; retrieval then costs one encoder pass and one index lookup.

2.2 Citation-derived training pairs

For a paper p with reference list $R(p)$, we treat two papers as related when they are cited together by at least m later papers. The positive set of a query paper p is

$$P(p) = \left\{ d \;:\; |C(p) \cap C(d)| \geq m \right\}, \tag{2}$$

where $C(\cdot)$ is the set of papers citing its argument. Hard negatives are drawn from the top BM25 results for the title of p, excluding any paper within two hops of p in the citation graph. This exclusion is what keeps lexically similar but genuinely related work out of the negative set.

2.3 Training objective

Given a batch of B queries, each with one positive d_i^+ and a set N_i of hard negatives, we minimise the contrastive loss

$$\mathcal{L}_{\mathrm{c}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{e^{s(q_i, d_i^+)}}{e^{s(q_i, d_i^+)} + \sum_{d \in N_i \cup D_i^-} e^{s(q_i, d)}}, \tag{3}$$

where D_i^- contains the positives of the other queries in the batch. We add a distillation term $\mathcal{L}_{\mathrm{kd}}$ that matches the score

Table 1 Retrieval quality on the three SciRank splits. nDCG@10 and Recall@100 in percent; latency in milliseconds per query. The best score in each column is set in bold.

	Bio		Phys		CS		
Model	nDCG@10	R@100	nDCG@10	R@100	nDCG@10	R@100	ms
BM25	38.2	71.4	41.0	74.9	36.5	69.8	18
Dense, web pairs	35.7	76.3	37.9	77.2	39.1	80.4	9
Dense, mined negatives	40.8	81.0	42.6	82.5	43.3	84.7	9
Late interaction	43.5	84.2	44.9	85.1	**46.8**	87.9	112
CADRE (ours)	**47.9**	**90.6**	**48.7**	**91.2**	46.2	**91.4**	9

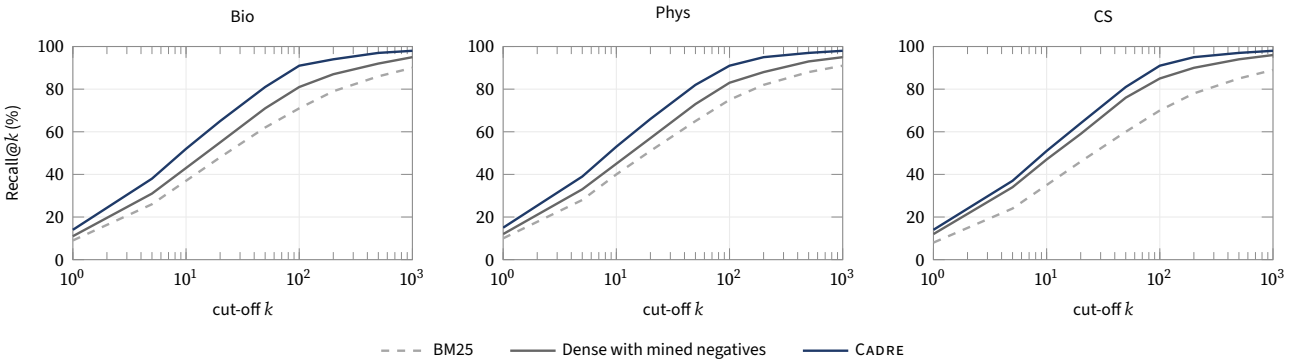


Figure 1 Recall as a function of the cut-off k on a logarithmic axis. The gap between CADRE and the lexical baseline is widest between $k = 20$ and $k = 100$, the range a reader or re-ranker actually inspects.

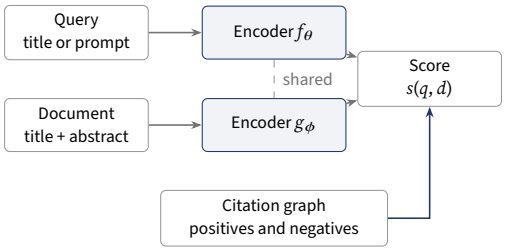


Figure 2 Training pipeline. Both encoders share weights; the citation graph supplies the positive and negative documents for each query.

distribution of a cross-encoder teacher on the same candidates, and train on $\mathcal{L} = \mathcal{L}_{\mathrm{c}} + \lambda\, \mathcal{L}_{\mathrm{kd}}$ with $\lambda = 0.5$. Figure 2 summarises the pipeline.

3 Experimental setup

Corpora. We pre-train on 9.6 million papers with their reference lists and evaluate on the three SciRank splits [7]: *Bio* (life sciences), *Phys* (physics and astronomy) and *CS* (computer science). Each query is a paragraph from a related-work section, and the relevant documents are the papers it cites.

Baselines. We compare against BM25, a general dense retriever trained on web questions [3], the same retriever with mined hard negatives [5], and a late-interaction model that stores one vector per token [9].

Metrics. We report nDCG@10 for the first page and Recall@100 for the candidate set a reader or a re-ranker would see. Latency is the median over 10 000 queries on one GPU, including the index lookup.

4 Results

Table 1 reports the main comparison. CADRE is the best model on five of the six quality columns and matches the latency of the other single-vector retrievers. Its largest gains are on Bio and Phys, where vocabulary differs most between subfields; on CS, the late-interaction model is 0.6 points ahead in nDCG@10 but answers twelve times more slowly and needs an index roughly forty times larger.

Figure 1 shows recall across cut-offs. At $k = 1$ all models are close, because the top result is usually the paper whose title repeats the query. The gap opens as k grows: the papers that CADRE recovers and BM25 misses are the ones phrased in another field's terms.

Ablation. Table 2 removes one component at a time on the Bio split. The two-hop exclusion in Equation (2) matters most: without it, a quarter of the mined negatives are papers the query actually cites, and the contrastive loss in Equation (3) pushes them away.

Failure cases. Most remaining errors involve very recent papers. A paper published after the citation snapshot has no co-citations, so its embedding is shaped only by its text; on queries whose relevant set is dominated by such papers, nDCG@10 drops to the level of the mined-negative baseline. Refreshing the graph monthly closes most of this gap at modest cost, since only the encoder for new documents

Table 2 Ablation on the Bio split. Each row removes one component from the full model.

Variant	nDCG@10	Δ
Full model	47.9	–
No two-hop exclusion	43.1	-4.8
No distillation ($\lambda = 0$)	45.6	-2.3
In-batch negatives only	44.0	-3.9
Co-citation threshold $m = 1$	46.4	-1.5

needs to run.

5 Conclusion

Citations are the most abundant relevance judgements in science, and they come for free. Using them to choose positives, and the citation graph to keep false negatives out of the training batch, gives a single-vector retriever that outperforms both lexical and general dense baselines on literature search without a latency cost. The obvious next step is to weight citations by the sentence they appear in, so that a citation of a method and a citation of a contradicting result no longer count the same.

Acknowledgements. We thank the maintainers of the open citation snapshots used in this work and the anonymous reviewers of an earlier draft for their comments.

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